

AllSERP: Exhaustive Per-Element Enrichment of the Versatile AdSERP Dataset

Andy Edmonds
andyed@gmail.com

Abstract

We release **AllSERP**, a typed AOI and per-element behavioral enrichment of the AdSERP commercial-intent SERP corpus [4]. AdSERP ships 2,776 trials of full-page screenshots, captured SERP HTML, 150 Hz Gazepoint eye tracking, evtrack mouse telemetry [5], scroll, and pupil signals against real Google SERPs collected before AI Overviews, but its bounding boxes cover only ad surfaces (15.5 % of attributable clicks). AllSERP adds pixel-accurate organic and widget bboxes via screenshot-anchored CV, semantic types across thirteen element types (nine main-axis) via an HTML parser, an inter-result gap-fill flavor (typed_gapfill), a cell-aware flavor that subdivides the top-ads carousel into its per-card cells, and X+Y click attribution that reaches 91.7 % of the corpus while flagging the rest at trial level. The typed ad-vs-non-ad partition is internally consistent with the shipped ad rectangles (0 disagreements across 38,250 classifications). We ship the pipeline, per-trial JSONs, a corpus CSV, and a browser-based replay viewer. Everything is reproducible from the AdSERP Zenodo volume. The release enables per-element click, fixation, regression, and above-fold analyses that the shipped ads-vs-organic split could not resolve.

CCS Concepts

• **Information systems** → **Users and interactive retrieval**; *Web log analysis*; • **Human-centered computing** → *User studies*.

Keywords

search engine results pages, multimodal datasets, eye tracking, mouse cursor tracking, area of interest, dataset enrichment

1 Introduction

AdSERP [4] is the only public IR dataset that combines eye gaze, cursor, scroll, pupil, and click telemetry against ground-truth screenshots and captured SERP HTML on commercial-intent Google SERPs at thousands-of-trials scale. Two early uses prove what the substrate carries. AdSight [7] predicts per-slot fixation time at NDCG = 96.07 ± 0.04 from cursor trajectories conditioned on slot metadata (slot type, normalized center position). Pupil-cognitive-load methods are being developed against AdSERP gaze on this corpus: real-time LF/HF [1] and per-fixation arousal via RIPA2 [3]. AllSERP raises the resolution at which work like these can run by replacing AdSight’s four-slot taxonomy with nine main-axis element types and by typing every main-axis card the user sees.

The shipped ad rectangles cover only ad surfaces: roughly **15.5 % of attributable clicks**. The remaining ~84.5 % land on organics, knowledge panels, image packs, People-Also-Ask widgets, and other untyped surfaces. Per-rank claims on AdSERP that pool ads with organics under “absolute rank,” or estimate organic geometry from h3 heading counts (which assume uniform result heights, though

actual layouts vary by 80–320 px), have inherited that gap. AllSERP closes it.

Contributions:

- (1) **Screenshot-anchored typed AOI extraction** with per-trial JSON output covering nine main-axis etypes (organic, dd_top, native_ad, top_places, knowledge_panel, paa, image_pack, other_widget, unknown_widget) plus chrome and off-axis surfaces, with a cell-aware superset flavor that subdivides the horizontal dd_top top-ads carousel into its per-card cells.
- (2) **Inter-result gap-fill and X+Y bbox-aware click attribution**. The gap-fill flavor extends adjacent organic bboxes via midpoint-split and uses X+Y containment, with a trial-level filter flagging off-axis clicks (right-rail ads, page chrome, far-off-target).
- (3) **Internal-consistency validation** against shipped ground truth: 38,250 ad-classification comparisons, 0 disagreements. Structural rather than independent-annotator (§3).
- (4) **Descriptive observed-behavior inventory** for nine main-axis element types: click share, fixation coverage, regression rate, above-fold incidence, on the full 2,776-trial corpus.
- (5) **Public release**. Single-script pipeline, per-trial JSONs, corpus CSV, and a browser-based replay viewer (§5). MIT code, CC-BY-4.0 derived data, matching the AdSERP corpus license.

2 Pipeline

2.1 Why screenshot-anchored

The AdSERP team’s published methodology extracted AOIs from the DOM at collection time [4], the cleanest source when the page is live. For *downstream reuse*, that route fails: re-rendering the saved 2022–2023 HTML in a 2026 browser produces layout drift of ~13 px median, growing to ~45 px at the page bottom, relative to the original screenshots (the source SERPs were captured on Chrome 110/Windows). Drift sources include missing external assets, Chrome version differences, font drift, and Google’s continual A/B-testing of SERP rendering. The shipped fixation and cursor streams are pixel-coordinated against the screenshots, so AllSERP anchors AOI geometry to those screenshots via CV and uses HTML only for structural labels. This complements the AdSERP team’s collection-time DOM approach: the DOM is cleanest at capture, and the screenshots are what survive for downstream reuse.

2.2 Four phases

Phase A produces card spans by per-row standard-deviation row-projection on the main column, with shipped ad rectangles taking precedence on overlap and a composite-trigger height triggering inner subdivision for image_pack / paa / top_stories. Phase B walks the saved HTML and assigns each card a type label via an 8-tier priority chain (heading text → structural class signatures →

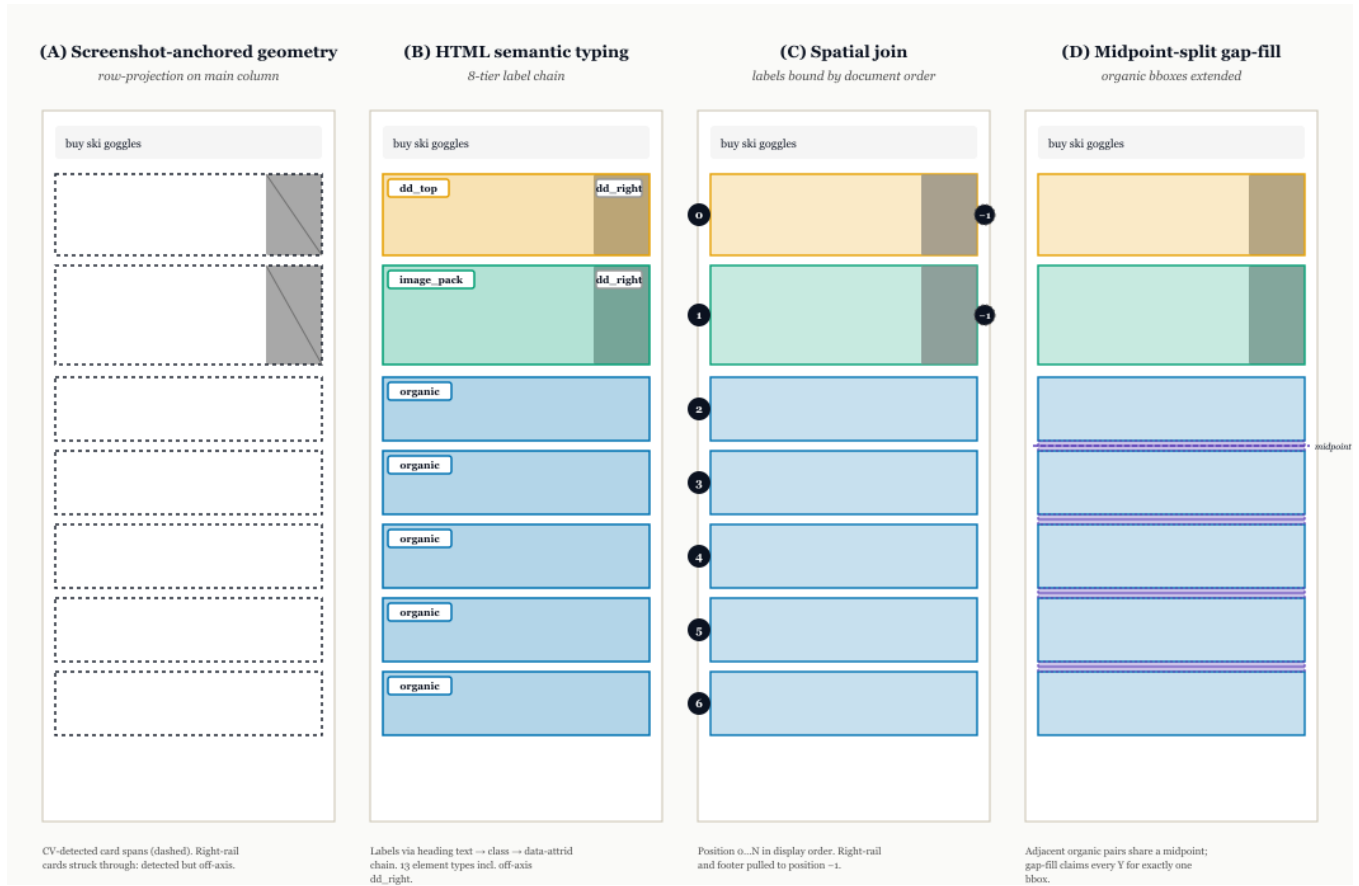


Figure 1: The four-phase pipeline applied to a synthetic SERP. (A) CV row-projection on the main column produces card spans (dashed outlines). Right-rail dd_right cards are detected but routed off-axis. (B) Each main-axis card receives a label from an 8-tier HTML chain. (C) Labels are bound to geometry by document order. Each main-axis card receives a position 0...N; right-rail and footer cards get position -1. (D) Adjacent organic pairs share a midpoint. The upper bbox extends down and the lower extends up, so every Y in the column belongs to one bbox. dd_top and image_pack are unchanged.

data-attrid $\rightarrow$ fallback structural patterns), producing 13 etypes including a chrome sweep for footer artifacts. Phase C binds labels to geometry by document order, the only stable axis when HTML and screenshot don't share a coordinate space. Phase D adds the typed_gapfill flavor by extending each pair of adjacent organic bboxes to their shared midpoint, clamped so an organic never crosses an ad or widget. Both flavors are released: the **tight-bbox flavor** (typed, Phase A $\rightarrow$ C) and the **gap-fill flavor** (typed_gapfill, Phase A $\rightarrow$ D). Analyses can pick the attribution semantics that match their use. A **cell-aware superset** (typed_gapfill_cellsplit) subdivides the horizontal dd_top carousel into its constituent cards, applying on the X axis the same midpoint-split Phase D applies on Y so a click in a card-gap attributes to the nearest cell; the same flavor carries sparse organic sub-cells and the off-axis dd_right right-rail block, the latter as a variance-reduction covariate rather than a modeling target (§5). Source code, parameter values, and per-trial outputs ship in the upstream repository (§5).

2.3 Click attribution and trial filter

Click attribution under the gap-fill flavor requires X and Y containment in a main-axis AOI bbox, with a small ± 5 X / ± 10 Y tolerance fallback for link-padding clicks. Y-only attribution would route right-rail ads, page chrome, and far-off-target clicks into adjacent main-axis AOIs that share their Y. X containment refuses those. The `is_main_axis_click(trial_id)` helper returns True iff a trial's final click lands in a main-axis AOI under this rule. In the AdSERP corpus, 231 trials are flagged at trial level: 67 dd_right (right-rail ad clicks), 91 page-chrome / search-tools / far-off-target, 73 with no clicks recorded or pathological click coordinates. Producers computing click-outcome features drop these trials.

3 Validation

The shipped ad rectangles [4] were extracted by the AdSERP authors against the same screenshots the gaze and cursor streams were recorded against, so they are pixel-anchored to the same truth source as our extraction. Phase C ad propagation matches the shipped

rectangles across all three ad etypes (dd_top, native_ad, dd_right) with **0 disagreements across 38,250 individual classifications**, mean IoU = 1.000, and 0 of 26,590 Phase A organic_result bboxes overlapping any shipped ad rectangle. We frame this as an **internal-consistency check** rather than independent-annotator validation: Phase C inherits ad identity by spatial overlap and is then checked against the same rectangles. The deeper non-ad partition (organic vs paa vs image_pack vs the rest) lacks an external label set. We validate it against the HTML structure that Phase B’s chain consumes and against visual spot-check on the curated 147-trial replay set. Independent-annotator validation on a held-out subset is named as future work.

Gaze-cursor spatial registration. Beyond the ad-rectangle internal-consistency check, the resource depends on gaze and cursor streams sharing a coordinate space. We probe this end-to-end: in the lead-up to the final click, did gaze ever land near the click target? For each trial with a final click ($n = 2,752$ with at least one fixation in the lead window), we take all gaze fixations whose midpoint falls in $[t_{click} - 1500 \text{ ms}, t_{click}]$ and compute the minimum Euclidean distance from any such fixation to the click coordinates. Median: **128.8 px**. Interquartile range: 80.4–206.5 px. p95: 446.2 px. Trials with a minimum gap exceeding 250 px ($\sim 3^\circ$ visual angle, past the GP3 HD calibration tolerance): **17.8 % (489)**. The concurrent at-click distance (the distance from the fixation that encompasses t_{click} to the click itself) is intentionally not the right metric here: gaze leads cursor by several hundred milliseconds on SERPs, so by t_{click} gaze has typically moved on, and the concurrent median is 506.8 px. The lead-window minimum is the spatial-registration probe. The concurrent median is reported only to disambiguate the two questions. Producer: `scripts/audit_gaze_cursor_coverage.py`. Full per-trial output and provenance metadata: `scripts/output/allserp/gaze_cursor_coverage.json`.

4 Element-Type Inventory

Population: 2,775 trials processed (1 trial dropped for missing meta or fixations), yielding 37,162 typed AOI rows under the gap-fill flavor. Table 1 reports the per-element-type behavioral inventory. Column denominators differ across columns (see notes below).

Column notes. *Fixated %* is over the etype’s full AOI population. *Click %* is over 2,634 click events attributed to a gap-fill AOI (89 % of all click events). The remaining 11 % are intermediate cursor clicks that landed off any main-axis AOI. The 91.7 % corpus-level rate quoted in the abstract counts trials whose *final* click lands main-axis under the trial filter, a different denominator. *Regressive %* is over the fixated subset and is therefore not directly comparable across etypes whose fixation rates differ widely (organic 56 %, dd_top 99.7 %). *Above-fold %* is the share of the 2,776 trials in which at least one AOI of that etype sits in the initial viewport.

Read. dd_top reaches near-universal fixation but only 9.6 % of clicks. The click-fixation dissociation is preserved at element-type granularity, sharper than at the ad-vs-organic level the original AdSERP analysis used. The pattern (36.4 % vs 5.9 % for native_ad, 99.7 % vs 9.6 % for dd_top) shows high fixation coverage decoupled from click share. dd_top’s near-universal fixation partly reflects its large, always-above-fold carousel geometry, so fixation coverage

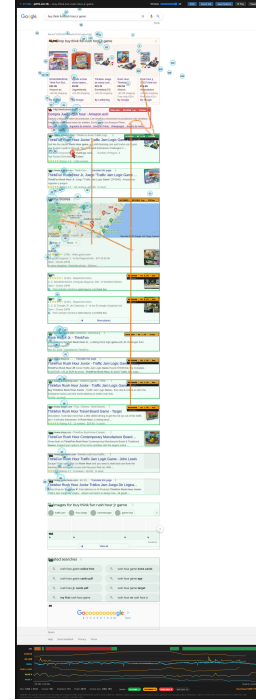


Figure 2: Replay viewer rendering of trial p010-b2-t6 (viewer ships with the repo, §5). Coloured rectangles: typed AOIs. Numbered circles: gaze fixations sized by dwell. Orange polylines: cursor trajectory. Sparklines: cursor speed, XY delta, pupil, LF/HF, gaze X/Y, AOI presence.

should be read alongside AOI area rather than as pure attentional pull. Organic results occupy the initial viewport on 97 % of trials and capture 79 % of clicks. Ads and widgets hold substantial above-fold geometry while drawing few clicks.

Within-carousel composition. The cell-aware flavor (§5.2) resolves the dd_top top-ads block into its cards. Of the 1,582 carousels, 1,550 (55.8 % of the 2,776 trials) subdivide into a median of four cells (Fig. 4, left; range 2–6, modal 4 at 61.9 %). Of the 237 *final* clicks landing inside a carousel (Table 1’s 254 dd_top clicks count all click events on the block; the cell split scores final clicks only), the leftmost cell draws 31.7 % on the modal four-cell layout ($n = 142$) against a 25 % uniform baseline, with the remaining cells near-even at 21–24 % (Fig. 4, right). Conditioned on the fixed four-cell layout the leftmost premium is modest, but pooling across carousel sizes and indexing by cell rank exposes a clean within-surface ordering: click-through declines monotonically from 4.3 % at the leftmost cell to 2.0 % at the fifth (Spearman $\rho = -1.0$ over the 231 clicks so ranked), with fixation count and dwell falling in step ($\rho = -0.94$ each). The block-level dd_top AOI cannot express either distribution: the subdivision is what makes within-surface position legible. Producers: `scripts/cellsplit_click_composition.py`, `scripts/compute_nb23_cellsplit_rank.py`.

Table 1: Per-element-type behavioral inventory under the gap-fill flavor. Etypes ordered by AOI count, descending.

etype	n_aois	fixated %	n_clicks	click %	regressive %	above-fold %
organic	22,346	55.6	2,084	79.1	57.8	97.7
native_ad	9,214	36.4	156	5.9	46.9	37.8
image_pack	1,584	52.0	55	2.1	59.3	21.0
dd_top	1,582	99.7	254	9.6	83.4	57.0
unknown_widget	788	17.4	7	0.3	26.3	0.0
paa	769	40.6	44	1.7	45.8	12.2
knowledge_panel	745	48.6	31	1.2	44.5	3.8
top_places	84	54.8	1	0.0	50.0	2.1
other_widget	50	58.0	2	0.1	44.8	1.1

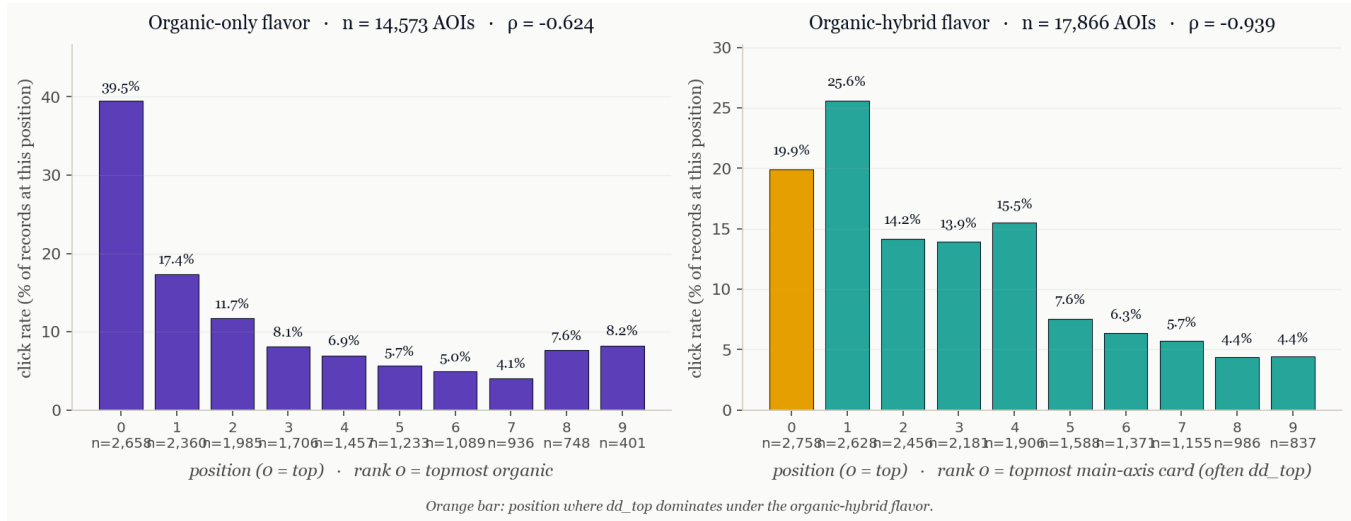


Figure 3: Click rate by SERP position under two position-numbering conventions, orthogonal to the bbox flavors of §2. Per-bar n is the number of AOIs at each position, and the panel header gives the total over positions 0–9. Click rate at each position is computed within that bar’s denominator only. Left, organic-only positions (positions index only organic results; 14,760 AOIs total, 74 % of the main-axis population). Position 0 is the topmost organic and the click heavyweight (39.5 %). Ranks decline to 4–8 % by position 7+ (Spearman $\rho = -0.624$). Right, all-main-axis positions (positions index every main-axis card, including dd_top ads and widgets; 19,908 AOIs). Position 0 is the topmost main-axis card, often a dd_top ad (orange bar, 19.9 %). The click peak shifts to position 1 (25.6 %), where the top organic now sits ($\rho = -0.939$). Consumers should pick the convention that matches their question: “what does the top organic earn?” versus “what does the topmost SERP slot earn?”.

5 Limitations and Public Release

5.1 Scope and provenance

Forced-choice task design. AdSERP elicits one click per query within a 1-minute window (extended to 2 minutes after a confirmation prompt) on a single query phrase per trial. The corpus does not sample query refinement, query reformulation, pagination, abandonment, multi-query sessions, or decision arcs longer than the forced-choice envelope. Behavioral statistics in §4 are properties of constrained-choice SERP evaluation, not of free Google search.

Pre-AI-Overviews snapshot. Data were collected between 2022-12-16 and 2023-03-13 (verified from per-trial entry timestamps), predating Google’s AI Overviews / SGE rollout (limited test May 2023, broad deployment May 2024) and Bard’s launch (March 21, 2023, three days after collection ended). The timing makes AllSERP

a clean pre-SGE baseline for commercial-intent Google SERP behavior, a fixed reference whose value grows as the live SERP diverges from it. Replications on post-May-2024 SERPs need an additional ai_overview etype, and the AI answer card has substantively different click-fixation dynamics. The four-phase pipeline shape is era-agnostic, and only Phase B’s label set needs to grow.

Phase D heuristic. The midpoint-split is a heuristic for the boundary between adjacent organic results, not DOM-derived ground truth. Misattribution is bounded by inter-result gap size (typically 5–60 px). We ship both flavors so consumers pick.

Top-ads cell subdivision. The cell-aware flavor (§5.2) subdivides the dd_top carousel into its per-card cells (§4), 100 % aligned to the block-level dd_top bboxes. The same flavor emits organic sub-cells, but their detection is preliminary: only 45.8 % of candidate

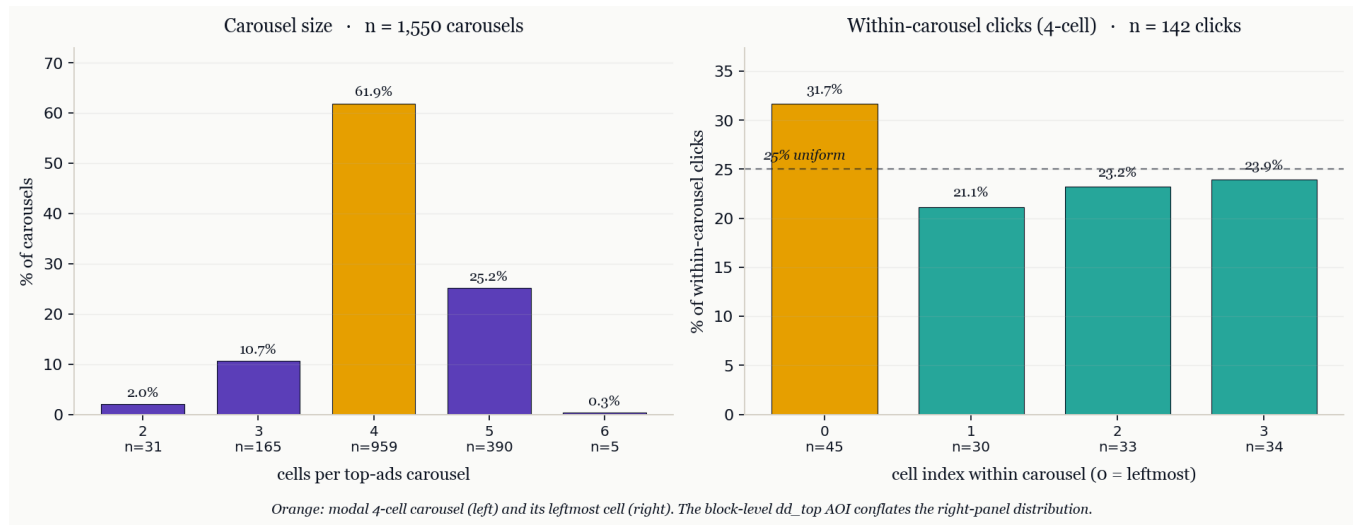


Figure 4: Anatomy of the `dd_top` cell split. Left: top-ads carousels hold 2–6 cards, modal 4 (61.9% of 1,550 carousels). Right: of the 237 clicks landing inside a carousel, the within-carousel distribution on the modal four-cell layout ($n = 142$) favors the leftmost cell (31.7% vs the 25% uniform baseline) but spreads across all cells. The block-level `dd_top` AOI conflates this distribution; the cell-aware flavor (§5.2) exposes it.

organic sub-cells align to a public organic result, so we ship the aligned subset (174 cells, 75 trials) and flag organic subdivision as the least-mature tier. Sub-card structure below the carousel (e.g. sitelinks within an organic) is not yet a stable surface.

Right-rail coverage. The pipeline routes shipped `dd_right` rectangles off-axis but does not recover right-rail organic-style results. About 1% of trials have a right-rail click with no shipped rectangle, a property of the original release.

Provenance. AdSERP collection followed the original team’s approved IRB [4]. AllSERP processes only de-identified telemetry and screenshots already released under CC-BY-4.0.

5.2 Public release

- **Code.** `attentional-foraging` (<https://github.com/and yed/attentional-foraging>) and `approach-retreat` (<https://github.com/andyed/approach-retreat>) under MIT license. Single-script entry point at `scripts/build_aois.py`.
- **Derived data (CC-BY-4.0).** The release ships three corpus-CSV flavors of per-AOI geometry plus matching per-trial JSONs (~19 MB across 2,776 trials) and a per-(trial, position) content-feature file with lexical, query-overlap, and semantic-cosine features:
 - `adserp_aois_by_trial_id_typed_gapfill.csv` – 9-etype gap-fill flavor (Phase A → D). **Recommended default** and the basis for the inventory in §4.
 - `adserp_aois_by_trial_id_organic_hybrid.csv` – 3-etype taxonomy (organic, native_ad, dd_top) with all-main-axis position numbering. Non-ad widgets (PAA, knowledge panels, image packs, top_places, other widgets) are pooled into organic, so the organic row count

(26,590) exceeds `typed_gapfill`’s (22,354; both are full-corpus CSV row counts over all 2,776 trials, whereas the §4 inventory tabulates the 2,775 processed trials). Use this flavor for continuous position-rank analyses where the topmost SERP slot is the unit of interest; use `typed_gapfill` when per-widget fixation or pupil signals must be disambiguated. Fig. 3 (right panel) uses this flavor.

- `adserp_aois_by_trial_id_typed_gapfill_cell_split.csv` – cell-aware superset of `typed_gapfill`. Filtering role == ‘parent’ and main_axis recovers `typed_gapfill` exactly; additional role == ‘cell’ rows subdivide the `dd_top` carousel (6,373 cells across 1,550 trials, 100% aligned to the block bboxes), carry sparse organic sub-cells (174 aligned cells, 75 trials), and expose the off-axis `dd_right` right-rail block (861 trials) and its cells (184). The right-rail rows (main_axis = false) are a **variance-reduction covariate**: most consumers filter them out, but conditioning on right-rail exposure absorbs variance in main-axis models. Use this flavor for within-surface (per-card) analysis of the top-ads block.

The repo README documents the cell-split methodology. One caveat carries across all flavors: snippet lexical diversity (type-token ratio) is correlated with rank position, so consumers should control for rank before claiming any per-position content effect.

- **Replay viewer.** <https://andyed.github.io/approach-retreat/replay/> renders typed AOIs on the source SERP screenshots for 147 curated trials.
- **Underlying corpus.** AllSERP requires the AdSERP Zenodo volume (<https://zenodo.org/records/15236546>). AdSERP authors release it CC-BY-4.0, and AllSERP does not redistribute screenshots or HTML.

Use. The corpus CSV is the join target for downstream click-rate, fixation, and content-feature analyses. Existing AdSERP signal files (cursor approach features, pupil power-ratio, saccade orientation) merge on `trial_id` and gain a per-AOI etype column without re-collecting data. The attentional-foraging README ships a minimal Polars worked example (corpus CSV $\rightarrow$ above-fold organic count; cursor-approach features $\rightarrow$ per-position click rate; per-trial JSON $\rightarrow$ typed-AOI inspection).

5.3 Citation

AllSERP ships only the typed AOI labels and the extraction pipeline. The shipped AOIs are pixel-anchored to AdSERP screenshots and indexed by AdSERP trial IDs, so any analysis that joins them to gaze, cursor, scroll, or pupil streams requires the AdSERP corpus, and both citations apply [4]. The only standalone use case for AllSERP is applying the pipeline to a different SERP corpus, in which case the AllSERP citation alone suffices.

6 Conclusion

AllSERP is a typed AOI and per-element behavioral enrichment of the AdSERP corpus. The pipeline anchors bbox geometry to the shipped screenshots, types every element via an 8-tier HTML parser, and fills inter-result Y gaps via midpoint-split. The typed ad partition is internally consistent with shipped ground truth across 38,250 classifications, and click attribution under X+Y containment reaches 91.7% of the corpus. The descriptive observations in §4 surface click-fixation dissociation, regressive return rates, and above-fold geometry at per-element granularity that the prior ad-vs-organic split did not resolve.

The substrate the enrichment unlocks is broader than the four metrics inventoried here. With per-element labels and geometry now joined to the multimodal telemetry the AdSERP team released, several lines of work become tractable on the existing 2,776-trial corpus without re-collecting data. Cursor approach-retreat episode geometry can be computed per AOI rather than per rank, characterizing the decision moment as a function of element type. The cell-aware flavor pushes this finer still: within-surface sampling—how many top-ads cards a user inspects before committing—becomes measurable per trial, a sub-trial signal the block-level `dd_top` AOI cannot express. Pupillometric and LF/HF cognitive-load signals [1, 3] can be conditioned on element type, separating the load profile of an organic-result read from that of a knowledge-panel or ad-card view. Content-feature analyses can be paired with element-type and behavioral outcomes to ask which content properties covary with engagement on which surfaces. Information-foraging theory [2, 6] provides a natural theoretical lens here, with query-snippet semantic cosine and query-token overlap as operational proxies for information scent at the per-AOI level.

What ships here is the geometric and typing layer those next steps depend on. The behavioral inventory in §4 is the descriptive baseline. The analyses it enables are the work the field can now do on a clean pre-AI-Overviews snapshot of commercial-intent SERP behavior, while the four-phase pipeline shape stays era-agnostic for replications on post-rollout SERPs.

7 AI Use Disclosure

This paper was drafted in collaboration with Anthropic’s Claude (Opus 4.7; the cell-split revision with Opus 4.8). Two automated audit tools checked the draft. `science-agent`¹ cross-referenced every citation against the bibliography and lit-notes, verifying that prose paraphrases matched each cited work’s actual claims. A second argument-rigor pass flagged ungrounded comparatives, causal language in descriptive prose, and numerical mismatches across sections, tables, and figure captions. The numbers in this paper come from a pipeline disciplined by stable Key-Claim identifiers in the upstream notebooks, which made mid-draft revisions cheap to recompute end-to-end without hand-typing or interpretation drift. Null findings are tracked alongside positive results in the released artifact. All quantitative claims, citations, and methodological decisions were verified by the authors, who take full responsibility for the content.

Acknowledgments

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¹<https://github.com/andyed/science-agent>